Change Detection on Remote Sensing Images Using a Dual-Branch Architecture With State Space Modeling and Graph-Based Local Interaction Network

Shujiang Luo, Aiye Shi, and Xin Wang

***Abstract*—In recent years, with the rapid development of remote sensing technology, the acquisition capability of high-resolution images has been significantly improved. However, change detection (CD) in complex scenes still faces multiple challenges. Traditional methods are often sensitive to seasonal variations and illumination fluctuations, which limits their ability to model global contextual dependencies and effectively suppress background noise. Moreover, local feature interaction mechanisms frequently fail to preserve subtle bitemporal differences due to inefficient cross-scale coupling, resulting in boundary blurring and false detections in multiscale target scenarios. In addition, many existing CD models rely on computationally expensive architectures, which hinders their practical deployment. To address these issues, we propose DSGNet, a dual-branch CD network that jointly exploits state space modeling and graph-based local interaction. The core design philosophy is to decouple global semantic dependency modeling from fine-grained local change representation and to integrate them efficiently through a guided attention-gated fusion strategy. The global branch introduces a directional state space model (DSSM), which extends Mamba and change state space models (CSSMs) to explicitly encode directional change trajectories across bitemporal images via state-driven dynamics, thereby enhancing the representation of large-scale structural variations. Meanwhile, the local branch incorporates a graph-based interaction module that dynamically constructs a learnable topological structure to facilitate explicit bitemporal feature alignment, significantly improving sensitivity to small-scale and subtle changes. In the fusion stage, a gated attention-guided feature fusion (GAFF) module adaptively balances global and local representations through a gating mechanism, which reinforces true change responses while effectively suppressing irrelevant background fluctuations. Experimental results demonstrate that DSGNet achieves superior performance on the WHU_CD, SYSU_CD, and DSIFN datasets, with $F1$-scores of 95.20%, 85.69%, and 77.39%, respectively.**

***Index Terms*—Change detection (CD), Mamba, remote sensing images, state space models (SSMs).**

I. Introduction

With the continuous advancement of remote sensing imaging technology, change detection (CD) has become an important research direction in the field of remote sensing. The objective of CD is to identify areas of surface changes by comparing remote sensing images of the same region acquired at different times. It plays a crucial role in understanding surface evolution and the impact of human activities, with wide applications in urban planning [1], [2], land use monitoring [3], disaster assessment [4], ecological and environmental protection [5], and natural resource management [6]. Especially with the increasing availability of high-resolution remote sensing imagery, CD has gradually transitioned from traditional heuristic algorithms to data-driven deep learning-based systems.

While higher spatial resolution brings rich texture and structural details for CD, it also leads to increased image heterogeneity and false differences, posing significant challenges in complex real-world scenarios. Traditional methods, such as algebraic difference [7], transformation analysis [8], and classification-based methods [9], achieved certain success in early applications. However, due to their heavy reliance on handcrafted features and rules, they struggle to generalize across diverse object types, change patterns, and environmental disturbances, resulting in poor robustness and generalization on high-resolution imagery. Therefore, it is imperative to adopt methods with stronger feature representation and modeling capabilities.

The advent of deep learning has revolutionized CD. Since Caye Daudt et al. [10] first introduced fully convolutional networks (FCNs) to CD tasks, and convolutional neural network (CNN)-based methods have gained rapid popularity. CNNs automatically learn local spatial features through multilayer convolutions and nonlinear activations, alleviating reliance on manual feature design. Zhang et al. [11] proposed DSIFN, which enhanced feature discrimination using deep supervision. Fang et al. [12] proposed SNUNet, which enhanced intertemporal interaction and fusion through dense connections. While these methods improved detection accuracy and robustness, the limited receptive field of CNNs prevents them from capturing long-range semantic relationships, leading to missed or false detections in scenarios with cross-scale and sparse changes. To further strengthen global contextual reasoning, subsequent works began exploring architectures beyond traditional convolutions.

To overcome this limitation, the Transformer model has emerged as a new research hotspot due to its self-attention mechanism that models arbitrary positional dependencies. Dosovitskiy et al. introduced [13] Vision Transformer (ViT),

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which segments images into patches and encodes positional information, enabling convolution-free image classification. In CD tasks, Chen et al. [14] proposed BIT, enhancing the modeling precision of intertemporal relationships. These methods expand the model's receptive field and capture large-scale semantic differences. However, the quadratic growth of Transformer's complexity with image size limits its efficiency in high-resolution and dense prediction tasks.

To reduce computation, researchers introduced various approximations; linear attention [15], local window attention [16], and sparse attention [17] have shown promising results. However, these often trade off global modeling capability, limiting performance gains. Therefore, efficiently modeling global dependencies with reduced complexity remains a key research focus. Recent studies on hybrid architectures offer promising directions for balancing local inductive bias and global reasoning. For instance, Yang et al. [18] propose an enhanced hybrid CNN–Transformer network that integrates local and global features through a dual-branch structure and innovatively incorporates symmetric frequency refinement modules (high-frequency feature transformation (HFFT)/band-pass frequency feature transformation (BFFT) and a kolmogorov-arnold network (KAN)-based attention mechanism to extract key change information in the frequency and semantic domains. In a complementary approach, Fiaz et al. [19] present a Siamese-based hybrid retentive network, whose core lies in a feature difference module that parallelizes depthwise convolution and multihead retention mechanisms, enhancing discriminative power for multiscale changes while reducing computational overhead.

Concurrently, spiking neural networks (SNNs), often regarded as the third generation of neural networks and inspired by biological brains, have garnered attention due to their event-driven nature and potential for extremely low power consumption on neuromorphic hardware. SNNs process information asynchronously via discrete spike trains and internal membrane potential dynamics, making them inherently suitable for data with intrinsic temporal characteristics. In remote sensing, SNN applications have expanded from scene classification to tasks like optical flow estimation. For instance, Wu et al. [20] successfully converted trained DNNs into SNNs for remote sensing scene classification, achieving high accuracy with significantly reduced energy consumption by introducing multibit spiking and channel normalization. Yang et al. [21] proposed a self-supervised SNN learning framework [self-supervised learning based on hierarchical information bottleneck (SeLHIB)] based on information bottleneck theory for robust optical flow estimation from event camera data, demonstrating superior generalization and over 90% energy reduction compared to artificial neural network (ANN) under noisy conditions. Recently, Duan et al. [22] further expanded the applicability of SNNs by introducing an online adaptive framework for remote sensing tasks, in which forward temporal optimization based on approximate backpropagation through time (BPTT), an adaptive activation scaling (AAS) strategy that maintains performance under low time steps, and a confidence-weighted mechanism for detection tasks collectively enable efficient unsupervised adaptation of pretrained SNNs on edge devices, achieving ANN-level competitiveness with orders-of-magnitude higher energy efficiency across classification, segmentation, and detection benchmarks. These works validate the feasibility and energy advantages of SNNs for remote sensing vision tasks. For CD specifically, SNNs' temporal processing capability and the sparsity of spike events theoretically make them well-suited for analyzing dynamic changes between multitemporal remote sensing images, particularly appealing for power-constrained onboard or edge computing. However, effectively adapting SNNs for pixel-level CD using bitemporal or multitemporal sequences—while overcoming training complexity, temporal error accumulation, and the performance gap with CNN/Transformer models—remains an area requiring further exploration.

In this context, the Mamba architecture [23] offers a novel paradigm for sequence modeling in vision tasks. Based on state space models (SSMs), Mamba introduces selective scanning and learnable dynamics to model long sequences with linear time complexity. After its success in natural language processing, Mamba was extended to vision. Zhao et al. [24] proposed RS-Mamba, combining state-space modeling with residuals for multiscale change modeling. Chen et al. [25] introduced ChangeMamba, integrating temporal modeling with feature alignment. Zhang et al. [26] proposed CDMamba, combining global modeling with local perception via bitemporal fusion. These advancements not only validate the effectiveness of SSM for CD but also demonstrate that SSM-based architectures provide a stable and fully differentiable alternative to biologically inspired temporal models, enabling efficient long-range modeling while maintaining optimization robustness and dense prediction accuracy.

Despite the success of Mamba-based models, they primarily focus on global modeling, neglecting fine-grained detail representation in change regions. Remote sensing changes often exhibit multiscale, localized, and complex morphology, and insufficient modeling of local structures and edges can result in blurred boundaries and misclassification. Furthermore, current Mamba models lack explicit interaction mechanisms between global and local features, hindering the extraction of consistent and contrasting semantics across bitemporal images. From this perspective, although SNN research highlights the importance of temporal sparsity and local responsiveness, directly adopting spike-based mechanisms for pixel-level CD remains challenging due to training complexity and dense prediction requirements, which further motivates the need for SSM-based global modeling complemented by explicit local structural modeling.

To address these limitations, we propose a dual-branch architecture for CD: a global branch built on SSM and self-attention to capture long-range dependencies and semantic structures, and a local branch with a lightweight graph-based module to simulate local structural features and enhance modeling of edges and subtle changes. This design aims to comprehensively capture both macrosemantic structures and microchange details across scales and classes.

Specifically, we adopt an improved change SSM (CSSM) [27] as the core of the global branch and introduce directional SSM (DSSM) to guide Mamba in perceiving structural

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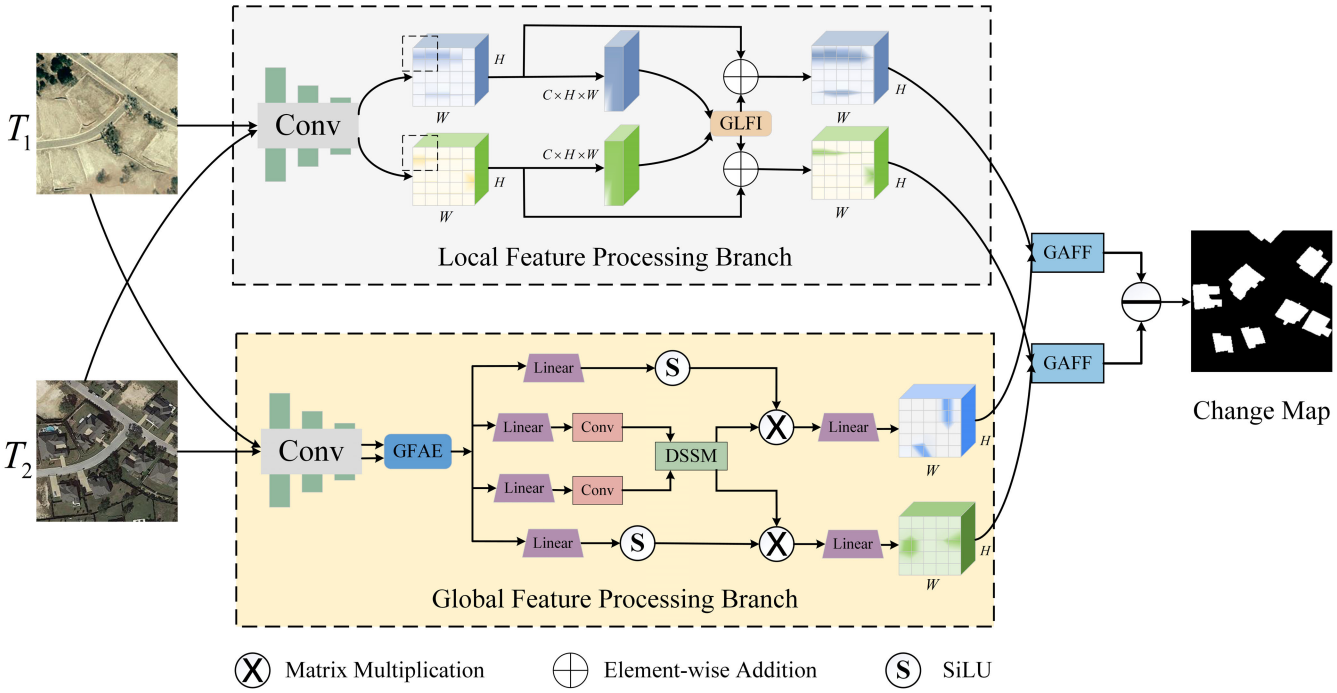


Fig. 1. Proposed DSGNet architecture.

changes from multiple directions, enabling effective semantic abstraction and difference modeling. The local module uses attention-based graph node representations to dynamically model local feature distributions, achieving refined local CD.

For efficient fusion, we design a GAFF module that leverages global semantic information to guide local feature distributions and uses a gating mechanism for adaptive weight adjustment, enabling collaborative global–local modeling. The overall architecture improves spatial adaptability and semantic discrimination while maintaining inference efficiency.

In summary, our work extends the advantages of deep models in CD by integrating long-range modeling via SSM and local representation via graph structures. The global–local fusion mechanism enables precise and comprehensive CD, showcasing promising practical potential.

We name this hybrid model DSGNet. Our main contributions are given as follows.

1) We propose a novel dual-branch framework for remote sensing image CD, with distinct processing paths for global and local features, achieving high accuracy, efficiency, and robustness.
2) Based on the Mamba architecture, we design a global feature fusion module and DSSM to effectively process global features from bitemporal images.
3) For local features, we propose a graph-based interaction module with learnable nodes and local window mechanisms, capturing local details with significantly reduced complexity.
4) We propose GAFF to complementarily fuse global and local features, using global information to guide local attention and a learnable gating mechanism to adaptively adjust fusion ratios, achieving efficient collaboration.

Extensive qualitative and quantitative experiments on the WHU-CD and SYSU-CD datasets demonstrate that our model achieves state-of-the-art results, validating its superiority and competitiveness.

II. Proposed Method

A. Overall Architecture

The overall architecture of our proposed DSGNet is illustrated in Fig. 1. It takes two input images representing pre- and post-event scenes and consists of two core branches: the global feature processing network and the local feature processing network. The processed features are then fused by two dedicated fusion modules, followed by pixel-level subtraction to generate the final change map.

The global branch first extracts global features using the global feature association extraction (GFAE) module, which retains semantic associations while providing attention-based intertemporal guidance. The resulting features are then fed into the DSSM module to filter out redundant information and extract relevant features. These are subsequently fused with local features to generate the final binary change prediction map.

In the local branch, bitemporal features are processed via a graph-based representation module to generate 32 graph nodes that embed local information. This localized information is then broadcast from these nodes to each query token to obtain local feature outputs. These are fused with global features and subtracted pixel-wise to yield the final prediction.

B. Global Feature Association Extraction Module

To enhance the modeling of cross-regional semantic information in remote sensing CD, we design a GFAE module

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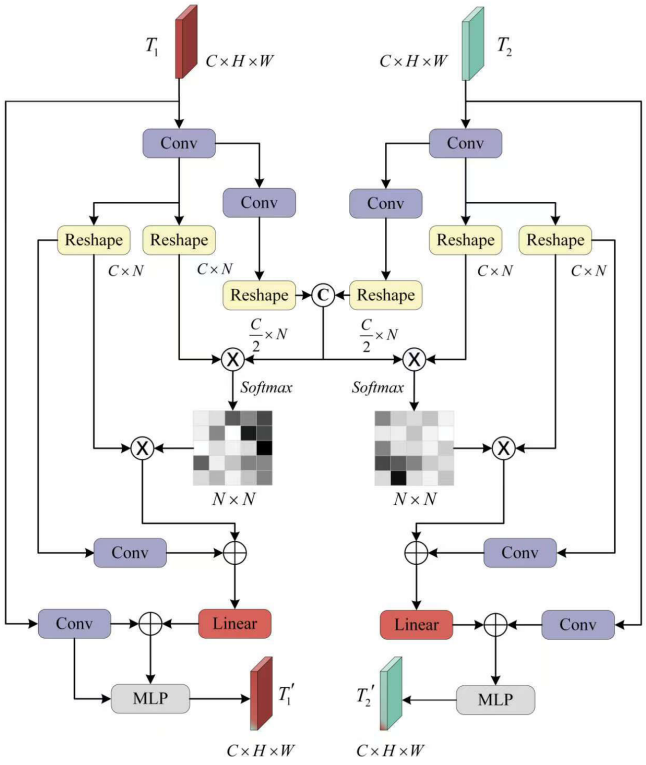


Fig. 2. Global feature association extraction module.

within the global branch, as shown in Fig. 2. This module aims to thoroughly mine the spatial semantic dependencies between bitemporal images to improve discrimination in change regions and provide more globally aware inputs for the subsequent state space modeling module.

In CD tasks, shallow features typically exhibit limited receptive fields. While they express edges and details well, they often lack understanding of the global context. This becomes particularly problematic for large-scale or structurally complex objects such as building clusters and road networks. Additionally, bitemporal images often suffer from illumination changes, viewpoint differences, and background disturbances. Solely relying on local convolution is insufficient to capture latent temporal differences.

To address these issues, the GFAE module first extracts initial semantic features using convolutional operations. Then, it structurally reorganizes the image features and projects them into a unified feature space. Subsequently, global interactions are modeled across the two temporal features by constructing correlation graphs over spatial locations. These graphs measure the semantic dependency between any two regions, enabling semantic sharing across spatial and temporal domains. This facilitates early establishment of global correspondence and enhances sensitivity to long-range changes.

Moreover, the module integrates residual connections and channel fusion strategies to preserve fine details while enhancing global context representation. This alleviates the degradation of shallow features in deeper networks. Finally, the fused features are passed into the SSM-based module to capture complex temporal dependencies and dynamic variations.

In summary, the GFAE module strengthens the model's global semantic representation in shallow layers and introduces global consistency constraints, mitigating the limitations of conventional convolution in modeling long-range dependencies.

C. Directional SSM

In this section, we propose DSSM, a direction-aware module for state space modeling, to further improve the modeling ability of Mamba-based architectures in CD tasks. DSSM draws from the classic SSM and its improved version CSSM, integrating intertemporal co-modeling and direction-sensitive difference mechanisms to focus on change regions and suppress redundant features.

The continuous form of the SSM is expressed as the following differential equations:

$$\begin{cases} h(t) = Ah(t-1) + Bx(t) \\ y(t) = Ch(t) \end{cases} \tag{1}$$

where $h(t)$ denotes the hidden state, $x(t)$ denotes the input, and $y(t)$ denotes the output. $A \in \mathbb{R}^{N\times N}$ is the evolution parameter, while $B \in \mathbb{R}^{N\times 1}$ and $C \in \mathbb{R}^{1\times N}$ are projection parameters. This model characterizes the dynamic driving relationship of the input signal on the system's state and output. To make it suitable for discrete neural network modeling, the Mamba model discretizes this formulation using the zero-order hold (ZOH) method. The discrete form can be expressed as follows:

$$\begin{cases} h(t) = \overline{A}h(t-1) + \overline{B}x(t) \\ y(t) = Ch(t) \end{cases} \tag{2}$$

where $\overline{A} = \exp(\Delta A)$, $\overline{B} = (\Delta A)^{-1}(\exp(\Delta A) - I)\Delta B$, and Δ denotes the discretization operator. This discrete SSM structure has been employed in models such as Mamba to model long-range dependencies with linear complexity.

In CSSM, a dual-input modeling mechanism is introduced, where input features from the current and previous temporal states, $x(t)$ and $x(t-1)$, are used to compute feature differences via transformation operators B and B'. The continuous form is given by

$$\begin{cases} h(t) = Ah(t-1) + |B'x'(t) - Bx(t)| \\ y(t) = Ch(t) + Dx(t) \\ y(t)' = C'h(t) + D'x'(t) \end{cases} \tag{3}$$

where the difference term $|B'x'(t) - Bx(t)|$, computed using the $L1$ distance, characterizes the salient changes between bitemporal inputs and serves to drive the hidden state updates. It responds only to regions with temporal differences, thereby enhancing the model's ability to perceive change regions. However, due to the absolute-value operation, this formulation is symmetric with respect to temporal order and therefore incapable of distinguishing whether the underlying features increase or decrease over time, i.e., it lacks directional sensitivity. Furthermore, the output terms $y(t)$ and $y(t')$ retain residual components $Dx(t)$ and $D'x'(t)$ from the original inputs, which enhances information flowthrough the direct pathway and improves model stability and representational capacity.

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To enable trainable modeling, CSSM applies ZOH discretization to the above continuous system, yielding the following discrete form:

$$\begin{cases} h(t) = \overline{A}h(t-1) + |\overline{B}'x'(t) - \overline{B}x(t)| \\ y(t) = Ch(t) + Dx(t) \\ y(t)' = C'h(t) + D'x'(t). \end{cases} \tag{4}$$

The core of this structure lies in utilizing the $L1$-based difference term $|\overline{B}'x'(t) - \overline{B}x(t)|$ to drive the evolution of the hidden state, activating responses only in change regions. This effectively filters out temporally consistent features and allows the model to focus on actual changes. Nevertheless, opposite temporal variations with identical magnitudes are treated equivalently, which limits the model's ability to capture asymmetric or trend-dependent changes.

In DSSM, we retain the dual-input state-driven structure of CSSM but introduce a direction-aware mechanism into the difference term. Specifically, instead of using the $L1$ norm, we define a directional difference based on the squared difference formulation

$$\delta(t) = \left(B'x' - Bx\right)\left(B'x' + Bx\right) \tag{5}$$

where B and B' are state-driving parameters linearly generated from the input features. Unlike the $L1$-based formulation, this directional difference preserves the sign information of temporal variation, enabling the model to distinguish increasing and decreasing trends in the state-driving forces. By decomposing the squared difference into a product of difference and sum terms, $\delta(t)$ jointly encodes both variation direction and activation intensity, embedding directional semantics into the state transition process.

The state update in DSSM is formulated as

$$\begin{cases} h(t) = Ah(t-1) + \delta(t) \\ y(t) = Ch(t) + Dx(t) \\ y(t)' = C'h(t) + D'x'(t) \end{cases} \tag{6}$$

where $\delta(t)$ is a state-based dynamic difference tensor that captures directional trends of change. It reflects not only the magnitude of temporal differences but also their directional evolution. As a result, DSSM can model asymmetric temporal dynamics and infer whether changes correspond to amplification, attenuation, or structural transformation, rather than merely detecting their existence. From a dynamical systems perspective, $\delta(t)$ acts as a second-order directional operator in the state space, amplifying consistent temporal trends while suppressing random fluctuations.

Similar to CSSM, DSSM adopts ZOH discretization to preserve stability and linear complexity. To handle long sequences, DSSM further employs a chunk-wise accumulated state modeling strategy, where long sequences are divided into multiple chunks and historical inputs are aggregated via a segment-wise exponentially decayed inner product, resulting in the final discrete state output

$$\begin{cases} h(t) = \overline{A}h(t-1) + \overline{\delta}(t) \\ y(t) = Ch(t) + Dx(t) \\ y(t)' = C'h(t) + D'x'(t). \end{cases} \tag{7}$$

Through the above modeling approach, DSSM effectively enhances the ability to model change regions in remote sensing images while maintaining a compact structure and computational efficiency. It is particularly well-suited for capturing complex change patterns characterized by directionality, local nonrigidity, or structural drift. These properties indicate that DSSM is not only effective for modeling bitemporal change signals but also possesses inherent structural advantages that may extend beyond the specific setting of CD.

Building on these observations, the direction-aware state-driven formulation of DSSM also demonstrates promising generalization potential for broader remote sensing applications. Its ability to capture directional and structurally dependent feature evolution in the latent state space aligns well with land-cover classification tasks requiring discrimination between spectrally similar but morphologically distinct classes. Likewise, the capacity of DSSM to model nonrigid, orientation-related, and deformation-sensitive variations suggests strong applicability in object detection scenarios, where robust feature encoding under geometric perturbations and multiangle observations is crucial. These characteristics indicate that DSSM is not only an effective change modeling module but also a transferable component capable of supporting more general remote sensing representation learning.

In summary, DSSM integrates the global modeling capability of SSM with direction-aware change representation. With the aforementioned generalization potential while preserving linear temporal complexity, it significantly improves the modeling of spatiotemporal structural changes in remote sensing imagery. By constructing a difference-driven dual-branch state space network, DSSM substantially boosts the accuracy and robustness of CD, making it especially suitable for remote sensing applications.

D. Graph-Based Local Feature Interaction Module

In the local feature processing network, to extract local features more accurately and efficiently, we propose a graph-based local feature interaction (GLFI) module that integrates geometric–semantic dual-domain constraints, as illustrated in Fig. 3. This module combines softmax attention and linear attention. Its core idea is to dynamically model the local feature distribution of remote sensing images through a learnable set of limited graph nodes [28] and broadcast these localized representations to each token. The final local feature output is obtained as shown in the following equation:

$$\begin{aligned} Q_1 &= \text{Softmax}\left(G(Q)_1 \cdot G_1^{\text{T}}\right) \cdot \text{Softmax}\left(G_2^{\text{T}} \cdot K_1\right) \cdot V_1 + T_1 \\ &= \phi_q(Q_1) \cdot \left(\phi_k(K_1)^{\text{T}} \cdot V_1\right) + T_1 \\ &= \text{LinearAtt}^{\text{eqk}}(Q_1, K_1, V_1) \end{aligned} \tag{8}$$

where $\phi(\cdot)$ is a mapping function that transforms Q and K into vectors with specific attributes. The vector sequences $K_1, V_1 \in \mathbb{R}^{N \times C}$ ($C \ll N$), where $N = H \times W$, in which H and W are the height and width of the features, respectively, and C denotes the channel dimension. Softmax is the activation function, T denotes the transpose operation, and LinearAtt denotes the linear attention. T_1 represents the original input, appended to the output via a skip connection.

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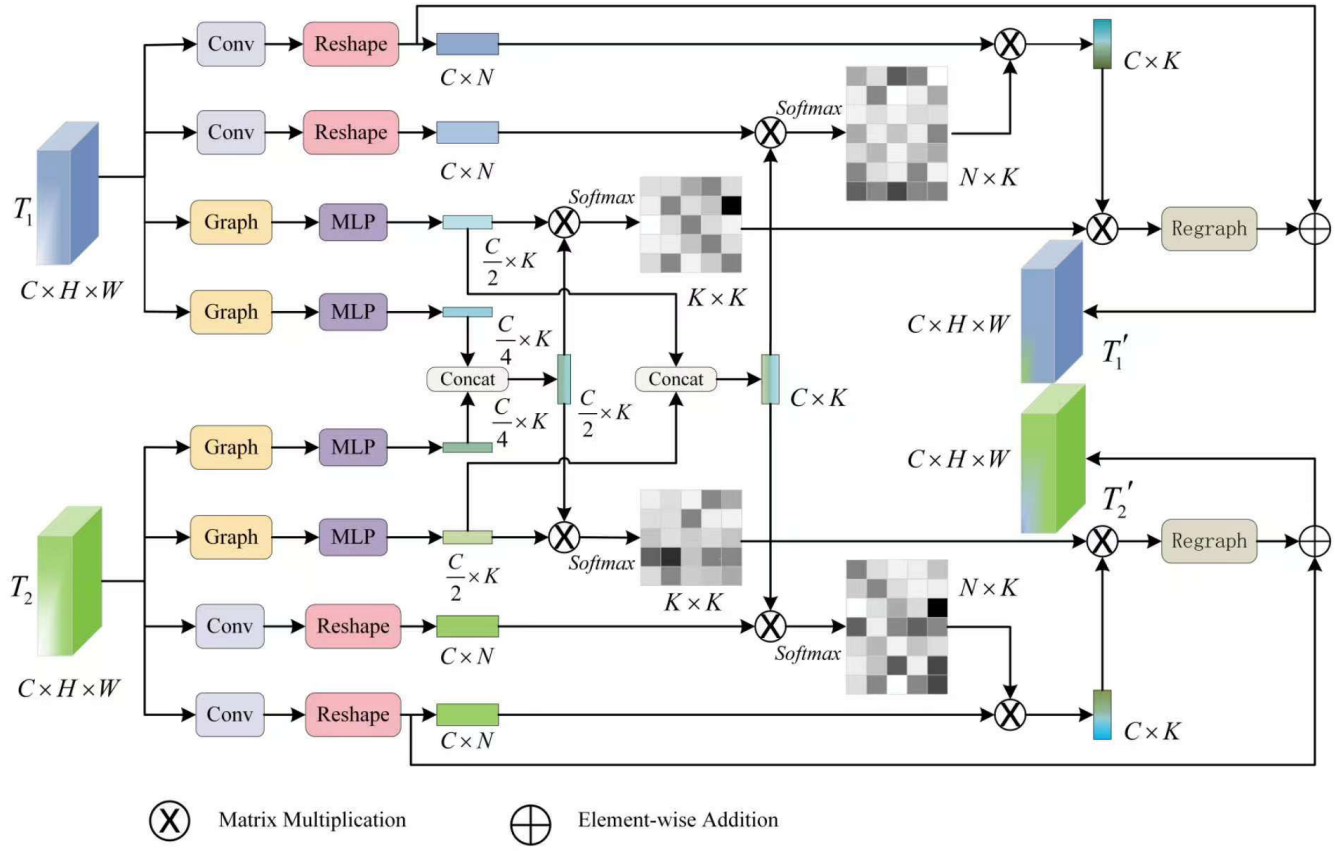


Fig. 3. GLFI module.

$G(Q)_1, G_1, G_2 \in \mathbb{R}^{N\times C}$ are local information proxies generated by the graph node network. For the input feature map $T \in \mathbb{R}^{B\times C\times H\times W}$, the graph node network first initializes several graph nodes $\{g_n\}_{n=1}^{N}$. Each node contains a feature anchor point $a_k \in \mathbb{R}^C$, representing the feature center of the kth $(0 < k < N)$ graph node, used to measure the feature similarity with pixels in the image; normalized spatial coordinates $l_k \in [0,1]^2$; a feature similarity scaling factor $\sigma_k^{\text{feat}} \in \mathbb{R}^C$; and a spatial constraint radius $\sigma_k^{\text{space}} \in \mathbb{R}$. These parameters, after being activated by the Sigmoid function, participate in the feature similarity measure and spatial proximity measure.

The feature similarity measure calculates the Euclidean distance between the pixel feature $x_i \in \mathbb{R}^C$ and the node anchor point

$$D_{\text{feat}}(p_i, g_k) = \frac{1}{2\left(\sigma_k^{\text{feat}}\right)^2}\left\|x_i - a_k\right\|_2^2$$
$$\sigma_k^{\text{feat}} = \sqrt{\frac{1}{C}\sum_{c=1}^{C}\left(\sigma_k^{\text{feat}}\right)^2} \tag{9}$$

where σ_k^{feat} denotes the average feature scale factor of the kth node, computed from the scaling factors along the node dimensions. Averaging over the channel dimension achieves cross-channel adaptive scaling, enhancing sensitivity to multispectral feature differences.

The spatial proximity measure establishes the geometric constraint between the pixel coordinate $p_i = (h_i, w_i)$ and the node position $l_k = (\tilde{l}_k^h, \tilde{l}_k^w)$

$$D_{\text{space}}(p_i, g_k) = \frac{1}{2\left(\sigma_k^{\text{space}}\right)^2}\left(\left(h_i - \tilde{l}_k^h\right)^2 + \left(w_i - \tilde{l}_k^w\right)^2\right). \tag{10}$$

Here, the normalized coordinates l_k are backprojected to the actual size $I_k = l_k \times (H-1, W-1)$ to preserve scale invariance. σ_k^{space} denotes the spatial scale factor of the kth node, which adjusts the size of its receptive field. This term reflects the spatial proximity between the node and the pixels and is used to guide the graph node in aggregating local features within its receptive field.

To balance computational efficiency and local consistency, a learnable local window constraint function is introduced

$$\mathcal{M}_k(p_i) = \mathbb{I}\left(\left|h_i - \tilde{l}_k^h\right| \le \frac{K-1}{2}\right)\cdot\mathbb{I}\left(\left|w_i - \tilde{l}_k^w\right| \le \frac{K-1}{2}\right) \tag{11}$$

where $\mathbb{I}(\cdot)$ is an indicator function that equals 1 if the condition is met, and 0 otherwise; K is the preset window size. This constraint limits the receptive field of each node to a local neighborhood, preventing the aggregation of globally redundant features and thereby enhancing regional sensitivity and directional consistency. At the same time, the spatial constraint strength can be dynamically adjusted via σ_k^{space}. The final

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pixel-to-node affinity is computed as

$$\alpha_{ki} = \frac{\exp\ -\ D_{\text{feat}}\left(p_i, g_k\right) - D_{\text{space}}\left(p_i, g_k\right) \cdot \mathcal{M}_k\left(p_i\right)}{\sum_{j=1}^{N} \exp\ -\ D_{\text{feat}}\left(p_i, g_j\right) - D_{\text{space}}\left(p_i, g_j\right) \mathcal{M}_j\left(p_i\right)} \tag{12}$$

where α_{ki} represents the soft association degree between pixel i and the kth graph node, serving as the soft assignment weight. This weight reflects the pixel's membership degree to the graph node in both spatial and feature domains. Finally, α_{ki} is aggregated with the flattened input features $T_{\text{flat}} \in \mathbb{R}^{B\times C\times HW}$ to obtain the final feature representation of the graph nodes

$$V_k = T_{\text{flat}} \cdot \alpha_{ki} \in \mathbb{R}^{B\times C\times N}. \tag{13}$$

$L2$ normalization is applied to the node features to eliminate magnitude bias caused by illumination variations, resulting in the final feature representation of the graph nodes, denoted as

$$\hat{V}_k = V_k / \|V_k\|_2. \tag{14}$$

The feature maps $G_1, G_2, G(Q)_1, G(Q)_2$, generated by the two graph node networks, broadcast feature information to the output features through a softmax attention operation. Subsequently, pixel-level feature results $X_k = \hat{V}_k \cdot \alpha$ are obtained via graph reprojection. Graph reprojection redistributes the high-level graph node features $\hat{V}_k$ back to the original image pixels, forming a dense representation X_k. This mechanism effectively restores spatial details while preserving the graph structural modeling capability, aligning with the requirements of image semantic processing. The dense representation is then fused with the input features to produce the final output.

In remote sensing image CD tasks, the local graph node (Graph) network enhances local feature processing capability through the following mechanisms. First, geometric–semantic dual-domain adaptation is achieved by employing a learnable spatial constraint radius σ_k^{space}, enabling nodes to adapt to scale variations of different land-cover objects, and by utilizing a feature scaling factor σ_k^{feat} to mitigate spectral shifts in multitemporal imagery. Second, dynamic receptive field control is realized through the combined effects of local window constraints and parameterized spatial metrics, which capture contextual information of critical regions while maintaining computational efficiency and suppressing irrelevant background interference. Finally, hierarchical feature representation is constructed via node feature aggregation, establishing an abstraction from pixel-level to region-level features and thereby enhancing the discriminative capability for local changes.

E. Gated Attention-Guided Feature Fusion Module

In remote sensing image CD and multiscale fusion tasks, local and global features exhibit a high degree of complementarity. Effectively integrating these two types of information—while preserving detailed resolution and enhancing semantic awareness—is key to improving model performance. However, conventional feature fusion strategies, such as feature concatenation, weighted summation, or symmetric attention-based cross-attention mechanisms, suffer from significant limitations: concatenation does not model information prioritization and results in a static fusion process; weighted averaging relies on fixed weights, lacking adaptability; and bidirectional attention mechanisms incur high computational complexity and often lead to overfitting or misalignment among multiresolution features.

To address these issues, this article proposes a GAFF module, as illustrated in Fig. 4. This module leverages global features to guide the distribution of local attention and employs a learnable gating mechanism to adaptively adjust the fusion ratio between the two feature types, thereby achieving more efficient collaborative modeling and fusion of local and global features.

This module takes local features $F_l \in \mathbb{R}^{B\times C\times H\times W}$ and global features $F_g \in \mathbb{R}^{B\times C\times H\times W}$ as inputs, generating fused features $F_f \in \mathbb{R}^{B\times C\times H\times W}$ through guided attention and gating mechanisms.

Specifically, global average pooling (AvgPool) and linear projection (Linear) are applied to F_l and F_g, respectively, to obtain their semantic representations. These representations are then multiplied with the original feature maps channel-wise, yielding enhanced representations F_l' and F_g' that augment the discriminative power of the original features.

Subsequently, F_l' and F_g' are concatenated along the channel dimension. A 1×1 convolution followed by a Sigmoid function generates a fusion gating weight map $W_{\text{gate}} \in [0,1]^{B\times C\times H\times W}$.

$$W_{\text{gate}} = \sigma\left(\text{Conv}_{1\times 1}\left(\text{Concat}\left(F_l', F_g'\right)\right)\right). \tag{15}$$

This gating weight map adaptively regulates the fusion ratio between local and global features. The fused feature is constructed as follows:

$$F_{\text{mix}} = W_{\text{gate}} \odot F_l' + \left(1 - W_{\text{gate}}\right) \odot F_g' \tag{16}$$

where $\odot$ denotes the element-wise multiplication. This process enables region-specific adaptive weighting that prioritizes local responses or emphasizes global semantics—a key design of this module. To enhance the representational capacity after fusion, the module finally concatenates F_{mix}, the locally enhanced feature F_l', the globally enhanced feature F_g', and the element-wise product $F_l' \odot F_g'$. This concatenated representation is processed through a nonlinear activation function [Gaussian error linear unit (GELU)] to produce the final fused output

$$F_f = \text{GELU}\left(\text{Concat}\left(F_{\text{mix}}, F_l', F_g', F_l' \odot F_g'\right)\right) \tag{17}$$

Compared to previous methods, the GAFF module guides the attention mechanism to regulate local information enhancement through global features, thereby strengthening the response of key regions. It incorporates a gating mechanism to learn fusion weights at the pixel level, offering greater adaptability and controllability. Moreover, the GAFF module features a lightweight overall architecture, relying solely on basic pooling, linear layers, and convolution operations, thereby avoiding complex matrix multiplications and high-dimensional tensor transformations. This makes it particularly suitable for feature fusion in high-resolution images. The introduction of this module significantly enhances our model's discriminative representation and boundary sensitivity in CD tasks, especially demonstrating superior fusion stability and task robustness in heterogeneous remote sensing imagery.

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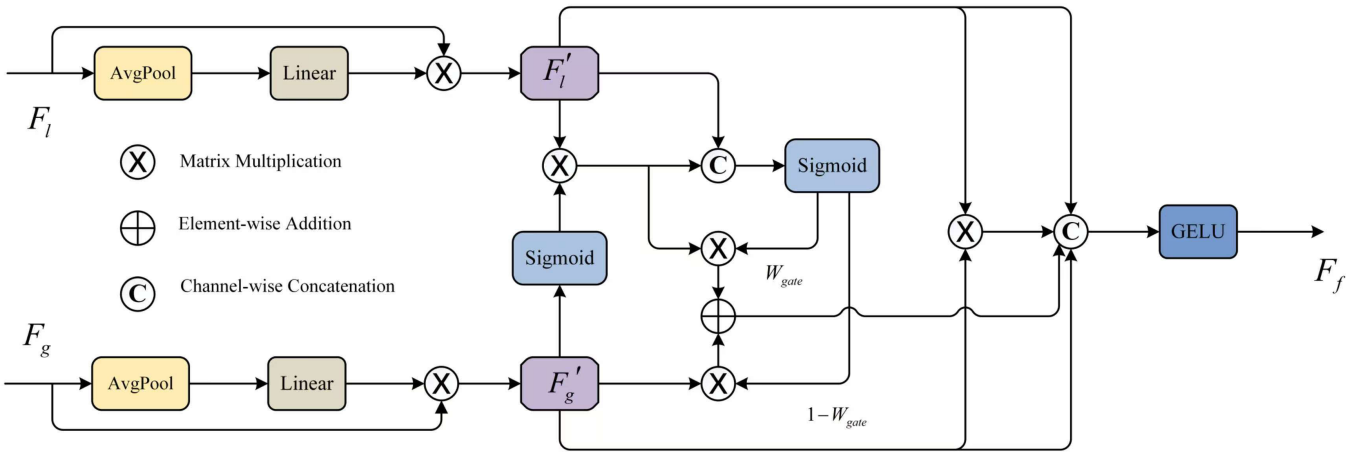


Fig. 4. Gated attention-guided feature fusion module.

III. Experimental Results and Analysis

A. Datasets

Extensive experiments were conducted on two typical CD datasets to validate the practical performance of our algorithm.

1) The WHU-CD dataset [29], released by Wuhan University, is a building CD dataset. It consists of a pair of aerial images with a spatial size of 32 507 × 15 354 and a resolution of 0.2 m/pixel. Since no official partitioning protocol is provided, we cropped the large image pairs into 256 × 256 patches and randomly split them into 6096/762/762 for training/validation/testing.
2) SYSU-CD dataset [30] from Sun Yat-sen University includes 20 000 image pairs with a resolution of 0.5 m/pixel and a size of 256 × 256. Among them, 8000/4000/8000 pairs are used for training/validation/testing. It covers buildings and many other types of objects, such as ships, roads, and vegetation, which presents a significant challenge for CD due to the large data volume and scene diversity.
3) DSIFN-CD dataset [11] is a publicly available CD dataset composed of images from six major cities in China with a spatial resolution of 2 m/pixel. The dataset contains various types of changes, including buildings, roads, and farmlands. The images are cropped into patches of size 512×512, resulting in 3600/340/48 pairs for training, validation, and testing, respectively.

B. Experimental Settings

1) Training Details: In our experiments, the number of graph nodes N was set to 32. We adopted the AdamW optimizer with weight decay and a polynomial learning rate decay strategy, where the initial learning rate was set to 0.0004. The entire training process ran for 200 epochs. All experiments were conducted on an NVIDIA 4070 Ti Super GPU. The loss function is a combination of cross-entropy loss and Dice loss [31]

$$L_{ce} = -\frac{1}{N}\sum_{i} y_i \log(\hat{y}_i) \tag{18}$$

$$L_{dice} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i} \tag{19}$$

$$L_{total} = \lambda_1 L_{ce} + \lambda_2 L_{dice} \tag{20}$$

where λ_1 and λ_2 are the loss weights, y_i is the ground truth value of the ith pixel, $\hat{y}_i$ is the predicted value of the ith pixel, and N is the number of pixels.

2) Evaluation Metrics: To evaluate the similarity between predicted change maps and ground truths, six metrics were used: precision, recall, $F1$-score ($F1$), intersection over union (IoU), and overall accuracy (OA), defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{21}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{22}$$

$$F1 = \frac{2}{\text{Recall}^{-1}+\text{Precision}^{-1}} \tag{23}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{24}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{25}$$

where TP, TN, FP, and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively.

3) Comparison Settings: To verify the effectiveness and efficiency of DSGNet in remote sensing CD, we compared it against several state-of-the-art models, including non-Mamba-based methods: FC-EF [10], SNUNet [12], DMINet [32], BiFA [33], SEIFNet [34], and Hyret-Change [19]; and Mamba-based methods: RS-Mamba [24], CDMamba [26], and CSSM [27]. All models were trained under the same conditions using official PyTorch implementations for fair comparison.

C. Performance Evaluation

1) Quantitative Results: Table I presents the overall performance of all comparison methods on the WHU-CD, SYSU-CD, and DSIFN test sets. The best results are highlighted in bold. Overall, the proposed method achieves

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TABLE I
QUANTITATIVE COMPARISON ON THREE BENCHMARK DATASETS, WITH THE BEST RESULTS HIGHLIGHTED IN BOLD. ALL RESULTS ARE EXPRESSED AS PERCENTAGES (%)

Models	WHU-CD					SYSU-CD					DSIFN				
	Pre.	Rec.	F1	IoU	OA	Pre.	Rec.	F1	IoU	OA	Pre.	Rec.	F1	IoU	OA
FC-EF [10]	91.31	89.36	90.31	83.56	98.56	88.89	81.06	84.04	73.60	89.60	75.60	71.93	73.51	61.55	86.35
SNUNet [12]	91.97	92.88	92.42	86.69	98.83	88.65	77.70	81.36	70.14	88.33	75.12	74.82	74.97	62.89	86.21
DMINet [33]	94.04	93.18	93.60	88.55	99.03	84.49	82.82	83.61	72.84	88.46	71.21	80.44	73.73	60.51	81.97
BiFA [34]	95.33	91.39	93.26	88.01	99.02	84.45	81.18	82.62	71.59	88.05	70.91	**80.51**	73.38	60.12	81.50
SEIFNet [35]	94.60	91.81	93.16	87.85	89.99	84.62	80.88	82.50	71.42	88.04	75.92	76.25	76.08	64.10	86.66
Hyret-Change [19]	94.58	94.13	94.36	89.78	99.14	84.86	80.00	82.01	70.81	70.90	–	–	–	–	–
CDMamba [26]	95.36	90.35	92.69	87.12	98.95	83.93	84.12	84.02	73.36	88.45	–	–	–	–	–
RS-Mamba [24]	93.73	94.00	94.84	90.58	99.23	86.14	83.88	84.92	74.70	89.46	73.24	76.98	74.82	62.39	84.89
CSSM [27]	94.43	93.79	94.10	89.36	99.11	85.81	**84.55**	85.15	75.00	89.48	74.69	77.76	76.05	63.86	85.88
DSGNet (Ours)	**95.73**	**94.69**	**95.20**	**91.18**	**99.28**	**87.09**	84.52	**85.69**	**75.81**	**90.04**	**78.70**	76.27	**77.39**	**65.76**	**87.99**

superior performance across all evaluation metrics, particularly demonstrating significant advantages in terms of $F1$ and IoU.

On the WHU-CD dataset, the proposed method attains an $F1$-score of 95.20%, surpassing RS-Mamba and CSSM by 0.36% and 1.10%, respectively. For the IoU metric, it reaches 91.18%, outperforming the best competing method, RS-Mamba (90.58%), with an improvement of 0.70%. Furthermore, compared to the best non-Mamba-based method, BiFA, the proposed method shows a substantial increase of 1.94% in $F1$-score, further indicating its superior precision and generalization capabilities in boundary delineation and semantic consistency. Notably, even when compared with recently proposed CNN-based and hybrid architectures such as SEIFNet and Hyret-Change, the proposed method consistently maintains a clear performance advantage, demonstrating its robustness under diverse architectural paradigms.

On the SYSU-CD dataset, which features complex scenes with diverse change targets such as buildings, roads, ships, and vegetation, the proposed method continues to lead. It achieves an $F1$-score of 85.69%, outperforming RS-Mamba and CSSM by 0.77% and 0.54%, respectively, and an IoU of 75.81%, exceeding CSSM by 0.81%. Compared with SEIFNet and Hyret-Change, which exhibit relatively limited performance under complex background interference, the proposed method demonstrates stronger discrimination capability and more stable segmentation results. These results demonstrate that the dual-branch architecture, which integrates global semantic modeling and local detail perception, plays a critical role in enhancing CD accuracy.

Furthermore, on the DSIFN dataset, which is characterized by large-scale scenes and fine-grained structural changes, the proposed DSGNet achieves the best performance across most evaluation metrics, attaining an $F1$-score of 77.39% and an IoU of 65.76%. This represents a clear improvement over all available comparison methods, validating the strong generalization ability of DSGNet on datasets with higher intraclass variability and complex spatial layouts.

In summary, the proposed DSGNet achieves state-of-the-art CD performance on all three benchmark datasets, further validating the comprehensive advantages of the dual-branch architecture in fusing spatiotemporal information and fine-grained features, while effectively suppressing pseudo-changes and background noise. This demonstrates its particular suitability for CD tasks involving complex surface scenes and cross-dataset generalization scenarios.

2) Qualitative Results: To further demonstrate the effectiveness of the proposed method, qualitative analyses are conducted on the WHU-CD and SYSU-CD test sets (Figs. 5 and 6). Additional qualitative results on the DSIFN dataset are presented in Fig. 7 to further evaluate the generalization capability of different methods under large-scale and complex scene variations. In these visualizations, different colors are used to distinguish the correctness of detection results: true positives (TPs) are shown in white, true negatives (TNs) in black, false positives (FPs) in red, and false negatives (FNs) in blue.

a) Visualization on WHU-CD: Several representative samples are selected for visual comparison. From the qualitative visual results, Fig. 5(a) and (b) depicts scenes with large-scale building changes. In these cases, most comparative methods can detect the main changed regions; however, they often suffer from blurred boundaries, missing details, or false responses. For example, FC-EF and SNUNet in Fig. 5(b) exhibit obvious missed detections and misclassifications in the roof areas, while CDMamba and RS-Mamba perform well in capturing the overall contour but still present certain structural misalignments and redundant responses in detailed regions. SEIFNet shows relatively complete change regions in Fig. 5(a) and (b), yet boundary noise and local misalignments are still observable, especially around complex building edges. In contrast, the proposed method not only faithfully reconstructs the spatial morphology of the changed regions at this large scale but also maintains higher precision along boundaries, showing strong consistency with the ground truth labels. In Fig. 5(b), several methods display expansion or contraction effects around building edges, whereas the proposed method achieves highly consistent localization and boundary preservation in change areas, fully reflecting its precise modeling capability of global spatial structures.

Fig. 5(c) and (d) focuses on the detection of small-scale changes under complex backgrounds. Such scenarios impose

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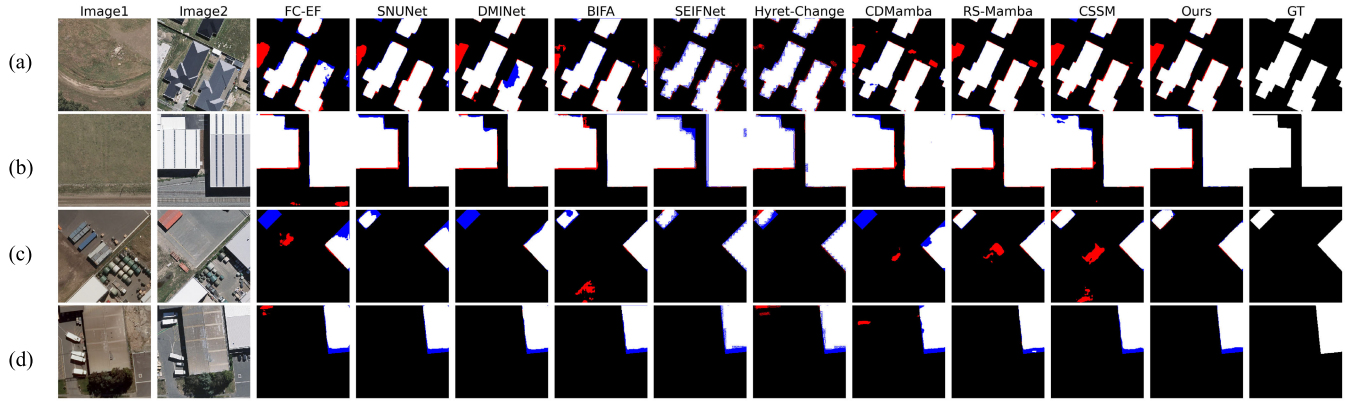


Fig. 5. Visualization results of different methods on the WHU-CD test set. (a)–(d) Representative samples. White indicates TPs, black indicates TNs, red indicates FPs, and blue indicates FNs.

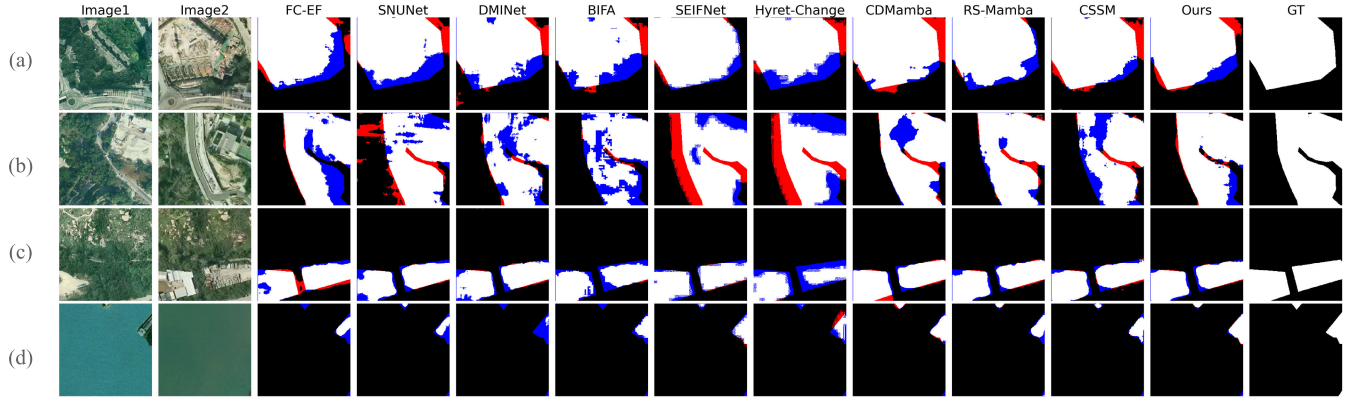


Fig. 6. Visualization results of different methods on the SYSU-CD test set. (a)–(d) Representative samples. White denotes TPs, black denotes TNs, red denotes FPs, and blue denotes FNs.

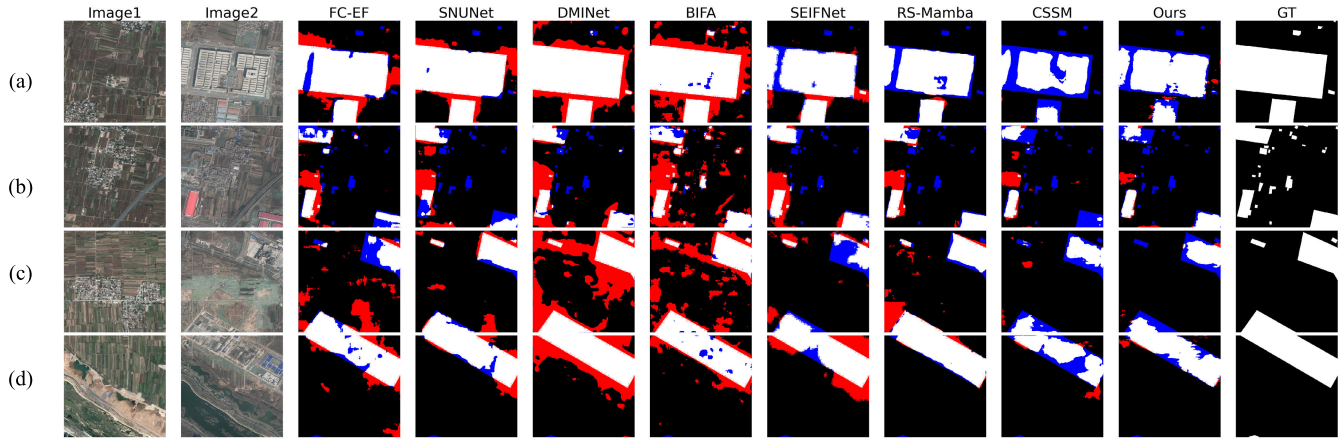


Fig. 7. Visualization results of different methods on the DSIFN test set. (a)–(d) Representative samples. White denotes TPs, black denotes TNs, red denotes FPs, and blue denotes FNs.

higher demands on the model's local perception and interference resistance capabilities. The results indicate that traditional methods like DMI-Net and BiFA exhibit numerous missed detections or false difference responses for small building changes. Although CSSM and CDMamba detect some changes in Fig. 5(c), they suffer from overdetection and background interference. Hyret-Change identifies partial change regions in Fig. 5(c); however, it also produces noticeable FPs in background areas, indicating limited robustness under complex scenes. By comparison, the proposed method demonstrates precise perception of subtle change regions in Fig. 5(c) and (d), accurately detecting target locations while effectively

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suppressing background noise, evidencing strong local detail modeling capacity.

In summary, the proposed method consistently maintains superior detection performance across change targets of varying scales and structures, owing to the globally local collaborative dual-branch architecture. The global branch enhances the modeling of large-scale changes by capturing cross-scale semantic dependencies, while the local branch strengthens adaptation to complex small targets through improved perception of local spatial structures and subtle differences. Under the guidance of a fusion mechanism, these two branches synergistically reinforce each other, exhibiting stronger robustness against irrelevant change interference, thereby endowing the model with enhanced robustness, accuracy, and generalization ability in CD tasks.

b) Visualization on SYSU-CD: Using a similar approach, several representative samples from the SYSU-CD dataset are selected for comparison.

Fig. 6(a) and (b) depicts typical large-scale building and road change scenarios, which often involve complex backgrounds, occlusions, and irregular structures, posing significant challenges to semantic understanding and boundary modeling. It can be observed that traditional methods such as FC-EF and SNUNet exhibit obvious overdetection and FP issues within these areas, particularly generating numerous spurious responses around road edges and building boundaries. Recent feature fusion methods, including SEIFNet and Hyret-Change, improve semantic consistency to some extent; however, boundary diffusion and redundant activations are still apparent in structurally complex regions. Although Mamba-based methods like CDMamba and RS-Mamba improve detection accuracy to some extent, redundant information and boundary blurring remain evident in detailed regions. In contrast, the proposed method accurately restores the overall contours of road and building change areas while suppressing interference from unchanged regions, demonstrating strong representational capability for large-scale, complex structural changes.

Fig. 6(c) presents a small-scale building CD scenario, where the smaller target size and often localized change regions impose higher demands on the model's fine-grained perception. The results reveal that most methods struggle to accurately identify the changed areas, with some models even completely missing the changes. This phenomenon is also observed in SEIFNet and Hyret-Change, which tend to produce fragmented or incomplete responses under small-scale structural changes. The proposed method, however, precisely detects the true change regions while minimizing background noise interference, indicating its advantage in local structure perception and detail preservation.

Fig. 6(d) illustrates a relatively extreme case of small-scale water surface changes, where most methods fail to detect changes due to the absence of clear structural edges. For this change type, the lack of explicit geometric cues significantly limits the sensitivity of most competing models, including recent fusion-based approaches. The proposed method nonetheless sensitively captures subtle change responses, exhibiting excellent change sensitivity and generalization ability. This phenomenon is likely attributed to the integration of local feature extraction capabilities, enhancing sensitivity to minor radiometric and textural variations.

c) Visualization on DSIFN-CD: To further validate the generalization capability of the proposed method, representative samples from the DSIFN-CD dataset are selected for qualitative comparison. Compared with WHU-CD, DSIFN-CD contains more diverse land-cover distributions and more complex background variations, which impose higher requirements on structural integrity modeling and robustness against background interference.

Fig. 7(a) and (b) illustrates scenes dominated by large-scale building changes. In these cases, most comparative methods can roughly identify the main changed regions; however, evident limitations remain. Specifically, FC-EF and SNUNet suffer from blurred boundaries and fragmented responses, leading to incomplete building structures. DMI-Net and BiFA show relatively better region completeness, yet they tend to introduce redundant FPs around building contours. SEIFNet and RS-Mamba capture the overall change extent more effectively but still exhibit local misalignments and internal noise within homogeneous building areas. In contrast, the proposed method generates compact and structurally consistent change maps, accurately preserving building morphology and sharply delineating boundaries, which are highly consistent with the ground truth.

Fig. 7(c) and (d) focuses on small-scale or irregular changes under complex backgrounds, where weak change signals and strong contextual interference coexist. The results show that traditional CNN-based methods, such as DMI-Net and BiFA, suffer from considerable missed detections, while CSSM and RS-Mamba tend to overdetect noisy background regions. Although SEIFNet can capture certain small-scale change targets, its performance is still affected by background variations. By contrast, the proposed method demonstrates superior sensitivity to subtle changes while effectively suppressing background noise, achieving accurate localization and clean boundaries across all samples.

Overall, the qualitative results on DSIFN-CD provide additional empirical evidence that the proposed method can effectively handle complex background variations and heterogeneous change patterns. In particular, the visual comparisons indicate that the proposed method maintains stable detection behavior across both large- and small-scale changes on DSIFN-CD, yielding clean boundaries and compact change regions without introducing excessive false alarms. These observations further demonstrate the practical effectiveness of the proposed framework on a newly introduced dataset.

Overall, these qualitative results demonstrate that the proposed dual-branch architecture, which combines global and local modeling, delivers stable and superior performance across CD tasks of varying types and scales. Specifically, the global branch contributes to a more reliable perception of large-scale man-made structures such as buildings and roads, while the local branch improves sensitivity to small-scale or weakly structured changes, including small buildings and water surfaces. The global branch enhances the model's understanding of large-scale structures and contextual relationships, facilitating the handling of structurally complex

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TABLE II
COMPARISON OF MODEL EFFICIENCY

Model	Param.(M)	FLOPs(G)
FC-EF	1.35	3.57
SNUNet	1.34	4.69
DMINet	6.24	14.51
BiFA	6.13	54.21
SEIFNet	27.91	8.73
Hyret-Change	48.37	46.43
CDMamba	7.66	9.81
RS-Mamba	49.98	21.93
CSSM	4.34	5.10
Ours	3.16	10.89

change regions. Meanwhile, the local branch strengthens the ability to capture subtle changes and small targets, effectively improving recognition of edges, local anomalies, and fine-scale differences. Through their complementary effects, the two branches jointly support consistent performance across different change types, further validating the effectiveness of the proposed modeling strategy in diverse remote sensing CD scenarios.

3) Model Efficiency: To further validate the efficiency of the proposed model, we compare the number of parameters (Params.) and floating-point operations (FLOPs) with several main methods, as shown in Table II. It can be observed that the proposed method achieves relatively low parameter count (3.16 M) and computational complexity (10.89 G) while maintaining competitive performance, demonstrating favorable efficiency among all nonminimal models.

Compared with representative lightweight models such as FC-EF (1.35 M and 3.57 G) and SNUNet (1.34 M and 4.69 G), although our method exhibits slightly higher parameter count and FLOPs, the substantially enhanced feature representation capability and change modeling capacity effectively compensate for this increase, resulting in superior overall detection performance.

In contrast, when compared with SEIFNet (27.91 M and 8.73 G), which relies on deeper network stacking to improve feature discrimination, the proposed method achieves a comparable computational complexity while significantly reducing the number of parameters. This demonstrates a higher parameter utilization efficiency and a more compact architectural design.

Compared with more complex architectures such as DMINet (6.24 M and 14.51 G) and BiFA (6.13 M and 54.21 G), the proposed method reduces both the parameter count and computational cost by more than half, thereby substantially improving runtime efficiency.

Particularly, compared with Hyret-Change (48.37 M and 46.43 G), which adopts a CNN–Transformer architecture to enhance long-range dependency modeling, the proposed method achieves significantly lower model complexity in terms of both parameters and FLOPs. This indicates that our model can effectively capture discriminative change information without relying on excessively heavy architectures.

TABLE III
ABLATION STUDY OF DIFFERENT MODULES

Models	WHU-CD
	Pre. / Rec. / F1 / IoU / OA
- GFAE	94.63 / 94.83 / 94.73 / 90.39 / 99.19
- Graph	94.99 / 94.14 / 94.56 / 90.12 / 99.18
- GAFF	86.69 / 93.22 / 89.65 / 82.59 / 98.28
- DSSM	90.94 / **94.98** / 92.85 / 87.36 / 98.86
Ours	**95.73** / 94.69/ **95.20** / **91.18** / **99.28**

TABLE IV
ABLATION STUDY OF DIFFERENT LOCAL WINDOW SIZES FOR GRAPH NODES

Size	WHU-CD
	Pre. / Rec. / F1 / IoU / OA
3×3	89.11 / 85.75 / 87.35 / 79.47 / 98.16
5×5	91.07 / 92.94 / 91.06 / 84.63 / 98.58
7×7	**95.73** / **94.69** / **95.20** / **91.18** / **99.28**
9×9	94.47 / 94.39 / 94.43 / 89.90 / 99.15
11×11	93.60 / 94.27 / 93.92 / 89.07 / 99.07

D. Ablation Study

In this section, we conduct a series of experiments on the WHU-CD dataset to investigate the impact of each component and parameter configuration within the proposed DSGNet on model performance, as shown in Tables II–VI.

1) Effect of Different Modules on Performance: To validate the effectiveness of key modules in our model, we design four sets of module-level ablation experiments. The results are summarized in Table III. Specifically, we either remove or replace certain modules within the model, including GFAE, Graph, GAFF, and DSSM, and use the complete model (Ours) as the performance benchmark. As the results indicate, the complete model achieves the best performance across all metrics on the WHU-CD dataset, with an $F1$-score of 95.20 and an IoU of 91.18—significantly outperforming all other configurations. This demonstrates the substantial contribution of the introduced modules to enhancing the model's discriminative capability and sensitivity to change regions.

Overall, these experiments quantitatively confirm the importance of each module within the model architecture. They also illustrate that the effective integration of key components—such as graph-based structural modeling, difference-guided feature fusion, and state space modeling—can significantly improve the overall performance of the model in CD tasks.

2) Impact of Different Local Window Sizes in Graph Nodes: To further investigate the effect of local window sizes in the graph network within the proposed BLEI module, we conduct ablation experiments on the WHU-CD dataset, as shown in Table IV. Specifically, we evaluate five different local window sizes: 3×3, 5×5, 7×7, 9×9, and 11×11. The results reveal that

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TABLE V
SENSITIVITY ANALYSIS OF KEY HYPERPARAMETERS ON THE WHU-CD DATASET

Learning rate	Loss weights		WHU-CD
	λ_1	λ_2	Pre. / Rec. / F1 / IoU / OA
4×10^{-4}	1	0	92.71 / 94.07 / 93.38 / 88.19 / 98.98
4×10^{-4}	0	1	92.71 / 95.13 / 92.71 / 87.13 / 98.83
4×10^{-4}	1	0.5	93.49 / **95.34** / 94.39 / 89.84 / 99.13
4×10^{-4}	0.5	1	95.08 / 94.96 / 95.02 / 90.87 / 99.24
4×10^{-4}	0.5	0.5	**95.73** / 94.69 / **95.20** / **91.18** / **99.28**
9×10^{-4}	0.5	0.5	81.79 / 92.07 / 86.13 / 77.80 / 97.54
1×10^{-4}	0.5	0.5	94.21 / 94.92 / 94.56 / 90.11 / 99.17
5×10^{-5}	0.5	0.5	93.72 / 94.38 / 94.01 / 89.27 / 99.09
1×10^{-5}	0.5	0.5	84.80 / 87.01 / 85.87 / 77.53 / 97.78

as the window size increases, the model performance initially improves and then degrades. The best detection performance is achieved when the window size is set to 7×7.

This indicates that a 7×7 window effectively captures local contextual information, facilitating accurate detection of fine-grained changes. When the window is too small, the model lacks sufficient capacity to perceive complex change patterns. Conversely, overly large windows introduce redundant information and noise, which compromise the model's ability to precisely localize change regions and lead to performance degradation.

Therefore, we adopt the 7×7 local window as the default configuration for Graph nodes in our model, striking a balance between structural awareness and overall performance. This experiment also highlights that in dense prediction tasks, the design of an appropriate receptive field plays a crucial role in ensuring model accuracy.

3) Sensitivity Analysis of Key Hyperparameters: To comprehensively evaluate the sensitivity of the proposed model to critical hyperparameters, we conduct a series of ablation experiments focusing on both the loss function weight coefficients and the learning rate on the WHU-CD dataset. Specifically, when analyzing the loss weight coefficients, the learning rate is fixed at 4×10^{-4}; conversely, during the learning rate sensitivity analysis, the loss weights are fixed as $\lambda_1 = \lambda_2 = 0.5$. The results are summarized in Table V, where λ_1 and λ_2 denote the weights of the cross-entropy loss and Dice loss, respectively.

The experimental results indicate that the model achieves optimal performance when the loss weights are balanced, which confirms the complementary nature of the cross-entropy loss and Dice loss in supervising both pixel-wise classification accuracy and region-level overlap consistency. In addition, the learning rate plays a critical role in determining the convergence behavior and stability of the training process. When the learning rate is excessively low, the optimization converges slowly and is prone to suboptimal local minima. Conversely, when the learning rate exceeds 1×10^{-3}, the training process becomes unstable, leading to degenerate predictions where all entries in the confusion matrix collapse to zero, rendering the model unable to perform effective learning.

A plausible explanation is that an overly large learning rate causes abrupt parameter updates that disrupt the delicate balance between the cross-entropy and Dice loss terms, resulting in gradient explosion or numerical instability. Under such conditions, the model tends to produce trivial predictions (e.g., predicting all pixels as a single class), which invalidates the computation of evaluation metrics and prevents meaningful training progress. Overall, a moderate learning rate of 4×10^{-4} yields the best convergence stability and overall performance across all evaluation metrics. Accordingly, we adopt $\lambda_1 = \lambda_2 = 0.5$ and a learning rate of 4×10^{-4} as the default hyperparameter configuration for the proposed model.

4) Robustness to Degraded Input Data: The robustness of the proposed method under degraded input conditions is further evaluated. During the inference phase, varying levels of Gaussian blur and Gaussian noise are introduced to simulate the impact of image quality degradation on CD performance. Fig. 8 illustrates the trends in $F1$-score for different models under degraded conditions on the WHU-CD dataset: the left subfigure shows the results under blur degradation and the right subfigure shows the results under noise degradation.

Regarding blur degradation, as the blur intensity σ increases, most models exhibit a gradual decline in $F1$-score, reflecting the adverse impact of spatial smoothing on boundary perception. Among them, FC-EF and CDMamba suffer the most pronounced performance degradation, dropping to 78.85% and 87.26%, respectively, when $\sigma = 1.5$. In contrast, the proposed DSGNet maintains an $F1$-score of 92.04% under the same condition, indicating superior robustness to blur and a strong ability to consistently perceive change regions even in heavily blurred images.

For noise degradation, the performance disparity among different methods becomes more evident as the noise level increases. While several models experience performance degradation to varying extents, some approaches exhibit severe instability under high noise conditions. In particular, HyretChange shows a dramatic performance collapse when the noise level exceeds 10%, with the $F1$-score dropping to 49.13% at 15% noise, indicating a high sensitivity to noise perturbations. In contrast, the proposed DSGNet demonstrates remarkable robustness, retaining an $F1$-score of 89.32% at a noise level of 15%, significantly outperforming BiFA (78.69%), CSSM (87.30%), and other competing methods. These results highlight the strong noise resistance and stable generalization capability of DSGNet under severe degradation scenarios.

5) Ablation Study on the Number of Graph Nodes in the GLFI Module: To evaluate the influence of the number of graph nodes in the proposed GLFI module, we conduct an ablation study by varying the number of learnable graph nodes while keeping all other settings unchanged. Since graph nodes serve as latent structural anchors for local feature aggregation and cross-temporal interaction, selecting an appropriate node count is crucial for balancing representational granularity and computational efficiency. We experiment with node counts of 8, 16, 32, 64, and 128 on the WHU-CD dataset.

The quantitative results are summarized in Table VI. When the number of nodes is small (e.g., 8 or 16), the model exhibits constrained representational capacity, leading to suboptimal $F1$-score and IoU score due to insufficient modeling of fine-grained spatial variations. Increasing the node count to 32

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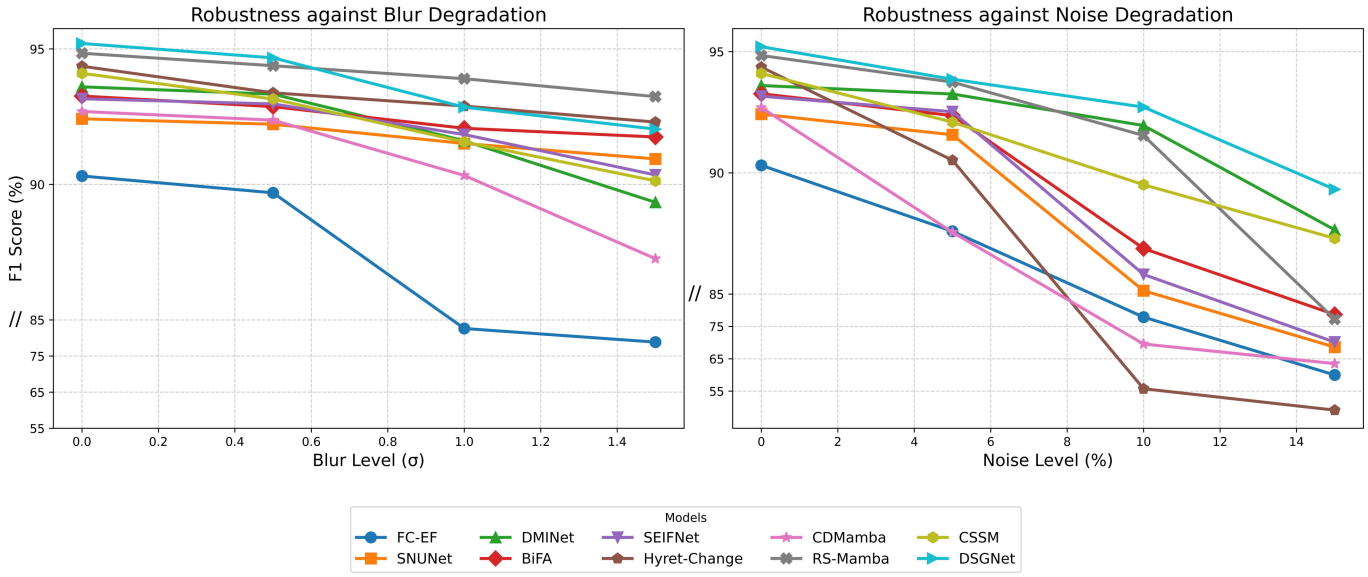


Fig. 8. Robustness performance of different models on degraded data in the WHU-CD dataset.

TABLE VI
ABLATION STUDY OF DIFFERENT GRAPH NODES

node	WHU-CD
	Pre. / Rec. / F1 / IoU / OA
8	94.17 / 94.63 / 94.40 / 89.84 / 99.14
16	92.61 / **95.14** / 93.83 / 88.92 / 99.03
32	**95.73** / 94.69 / **95.20** / **91.18** / **99.28**
64	94.88 / 94.09 / 94.48 / 89.98 / 99.17
128	92.32 / 94.59 / 93.42 / 88.26 / 98.97

yields the best overall performance, achieving an $F1$-score of 95.20% and an IoU of 91.18%, indicating that 32 nodes provide an optimal tradeoff between structural richness and computational overhead. Further increasing the node count to 64 or 128 does not improve performance; instead, a slight degradation is observed, likely caused by overfragmentation of local topology and redundant interactions that hinder stable optimization.

Overall, the results demonstrate that the default configuration of 32 graph nodes in GLFI is well justified, offering an effective balance between accuracy, robustness, and computational efficiency.

IV. CONCLUSION

To address the limitations of insufficient global modeling capability and weak local detail representation in remote sensing image CD, this article proposes a dual-branch CD network, DSGNet, which integrates comprehensive and local feature processing. The proposed method introduces a DSSM based on state space evolution, effectively enhancing the model's global perception of semantic change trends across multitemporal images. Meanwhile, a GLFI module and a local graph node modeling mechanism are designed to enable fine-grained modeling of small-scale change targets and facilitate cross-temporal feature interaction.

Moreover, the direction-aware state space modeling paradigm introduced by DSSM implies broader cross-task applicability. Its capability to encode directional structural variations and temporal evolution patterns provides a principled modeling advantage for other remote sensing tasks such as land-cover classification and object detection, in which structural discrimination, geometric robustness, and invariance to viewpoint changes are essential. Future research will further evaluate the generalizability of DSSM in these downstream tasks and explore its integration with task-specific architectures to build a unified and transferable state space-driven representation framework.

In addition, the GAFF module further improves the collaborative representation between global and local features. Extensive experiments on two representative datasets, WHU-CD and SYSU-CD, demonstrate that the proposed method achieves significant improvements in multiple metrics, including $F1$ and IoU. Overall, the proposed method exhibits clear architectural advantages, achieving a well-balanced tradeoff between accuracy and efficiency, and validating its feasibility and practical potential in remote sensing CD tasks. In future work, we plan to extend the proposed framework beyond multispectral inputs toward more heterogeneous multisource remote sensing scenarios. A primary direction is the integration of synthetic aperture radar (SAR) and hyperspectral image (HSI) data, which provide strong complementary characteristics to optical imagery. To this end, we will investigate: 1) a unified and modality-aware feature encoding architecture capable of accommodating the substantial radiometric, geometric and physical discrepancies among SAR, HSI, and optical domains; and 2) an adaptive cross-modal interaction mechanism that can dynamically balance and fuse complementary cues—such as the fine-grained spectral signatures of HSI, the all-weather scattering characteristics of SAR, and the spatial–structural information derived from optical imagery. Furthermore, we will explore how the proposed SSM-based global modeling and graph-driven local structural modeling

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can be extended to cross-modal spaces, enabling consistent long-range semantic reasoning and fine-grained spatial alignment across disparate sensing modalities. Ultimately, we aim to build a more generalizable, robust, and deployable CD framework capable of operating reliably under cloud cover, low illumination, seasonal variability, and other challenging conditions.

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